Functions and Function Notation

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Reading function graphs**

The graph of $f(x) = x^2 - 2$ is shown. Use it to estimate $f(1)$, and state the interval(s) where f is increasing.

From the graph, $f(1) = 1^2 - 2 = -1$. The parabola is increasing for $x > 0$.

2 **[Foundational]** • **Target: Average rate of change from function notation**

A ball's height is modeled by $h(t) = -5t^2 + 20t$ (meters, seconds). Find the average rate of change of height from $t = 1$ to $t = 2$, and interpret the sign of your answer.

$\frac{h(2) - h(1)}{2 - 1} = \frac{20 - 15}{1} = 5$. Since the average rate of change is positive, the ball's height is increasing on average over this interval.

3 **[Foundational]** • **Target: Function notation**

Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:

a $f(0) = 3$ **b** $f(2) = 1$ **c** $f(-1) = 10$ **d** $f\left(\frac{1}{2}\right) = 1$

(a) $= 3$; (b) $= 1$; (c) $= 10$; (d) $= 1$

4 **[Foundational]** • **Target: Function notation**

Consider the function $g(x) = \sqrt{x + 5}$. Evaluate:

a $g(4) = 3$ **b** $g(-1) = 2$ **c** $g(11) = 4$ **d** $g(-5) = 0$

(a) $= 3$; (b) $= 2$; (c) $= 4$; (d) $= 0$

5 **[Foundational]** • **Target: Function notation**

Consider the function $h(x) = \frac{x - 3}{x + 2}$. Evaluate:

a $h(0) = -\frac{3}{2}$ **b** $h(3) = 0$ **c** $h(-1) = -4$ **d** $h(5) = \frac{2}{7}$

(a) $= -\frac{3}{2}$; (b) $= 0$; (c) $= -4$; (d) $= \frac{2}{7}$

6 **[Foundational]** • **Target: Function notation**

If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a.

$3a - 7 = 11$, so $3a = 18$ and $a = 6$.

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7 [Foundational] • Target: Identifying functions

Determine whether each equation defines y as a function of x:

a $y = 2x + 1$ Yes — one output for each input.

b $x = y^2$ No — e.g. $x = 4$ gives $y = 2$ or $y = -2$.

c $y = |x|$ Yes — one output for each input.

d $x^2 + y^2 = 9$ No — a circle fails the vertical line test.

(a) Yes — one output for each input.; (b) No — e.g. $x = 4$ gives $y = 2$ or $y = -2$.; (c) Yes — one output for each input.; (d) No — a circle fails the vertical line test.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Identifying functions

Explain how the vertical line test is used to decide whether a graph represents a function.

If every vertical line crosses the graph at most once, each input x has at most one output y, so the graph represents a function. If any vertical line crosses the graph more than once, it is not a function.

9 [Intermediate] • Target: Combining functions

Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find:

a $(f + g)(x) = x^2 + 2x - 2$

b $(f - g)(x) = x^2 - 2x + 4$

c $(f \cdot g)(2) = 5 \cdot 1 = 5$

d $(f \circ g)(1) = f(-1) = 2$

(a) $= x^2 + 2x - 2$; (b) $= x^2 - 2x + 4$; (c) $= 5 \cdot 1 = 5$; (d) $= f(-1) = 2$

10 [Intermediate] • Target: Combining functions

A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.

a Interpret the value $C(0)$ in context. $C(0) = 250$: the registration fee alone, before any tutoring hours.

b Find $C(12)$. $C(12) = 250 + 540 = 790$ riyals.

c Solve $C(n) = 925$ and interpret your answer. $45n = 675$, so $n = 15$: a total cost of 925 riyals buys 15 hours of tutoring.

(a) $C(0) = 250$: the registration fee alone, before any tutoring hours.; (b) $C(12) = 250 + 540 = 790$ riyals.; (c) $45n = 675$, so $n = 15$: a total cost of 925 riyals buys 15 hours of tutoring.

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11 [Intermediate] • Target: Combining functions

Let $f(x) = \sqrt{x}$ and $g(x) = x - 4$. Find the domain of $(f \circ g)(x)$.

$(f \circ g)(x) = \sqrt{x - 4}$, which requires $x - 4 \geq 0$: domain $x \geq 4$.

12 [Intermediate] • Target: Piecewise-defined functions

A function is defined by $f(x) = x + 4$ for $x < 1$, and $f(x) = 2x + 1$ for $x \geq 1$. Evaluate:

a $f(-3) = 1$ **b** $f(0) = 4$ **c** $f(1) = 3$ **d** $f(4) = 9$

(a) $= 1$; (b) $= 4$; (c) $= 3$; (d) $= 9$

13 [Intermediate] • Target: Piecewise-defined functions

A phone plan charges a flat 60 riyals for up to 5 GB of data, then 15 riyals per additional GB beyond 5 GB.

a Write a piecewise formula $C(x)$ for the monthly cost, where x is data used in GB. $C(x) = 60$ for $0 \leq x \leq 5$, and $C(x) = 60 + 15(x - 5)$ for $x > 5$.

b Find $C(3)$ and $C(8)$. $C(3) = 60$ riyals; $C(8) = 60 + 15(3) = 105$ riyals.

c Is C continuous at $x = 5$? Explain. Yes — both pieces give $C(5) = 60$, so there is no jump at $x = 5$.

(a) $C(x) = 60$ for $0 \leq x \leq 5$, and $C(x) = 60 + 15(x - 5)$ for $x > 5$.; (b) $C(3) = 60$ riyals; $C(8) = 60 + 15(3) = 105$ riyals.; (c) Yes — both pieces give $C(5) = 60$, so there is no jump at $x = 5$.

14 [Intermediate] • Target: Piecewise-defined functions

Sketch-reasoning: the graph of $f(x) = |x - 2|$ can be written as a piecewise function without absolute value bars. Write it as such, and state $f(-1)$ and $f(5)$.

$f(x) = -(x - 2) = 2 - x$ for $x < 2$, and $f(x) = x - 2$ for $x \geq 2$. $f(-1) = 3$; $f(5) = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Even, odd, and one-to-one functions • Cross-Lesson: 5.02 The Dot Product

Determine whether each function is even, odd, or neither:

a $f(x) = x^4 - 3x^2$ Even — $f(-x) = f(x)$.

b $f(x) = x^3 - x$ Odd — $f(-x) = -f(x)$.

c $f(x) = x^2 + 2x$ Neither — matches neither pattern (e.g. $f(-1) = -1$, $f(1) = 3$, and $-f(1) = -3$).

(a) Even — $f(-x) = f(x)$.; (b) Odd — $f(-x) = -f(x)$.; (c) Neither — matches neither pattern (e.g. $f(-1) = -1$, $f(1) = 3$, and $-f(1) = -3$).

16 [Challenging] • Target: Even, odd, and one-to-one functions • Cross-Lesson: 5.03 Parametric Equations

Determine whether each function is one-to-one (passes the horizontal line test). If not, give a horizontal line that crosses the graph twice.

a $f(x) = 2x + 5$ One-to-one — every horizontal line meets the graph at most once.

b $f(x) = x^2$ Not one-to-one — e.g. $y = 4$ meets the graph at $x = \pm 2$.

(a) One-to-one — every horizontal line meets the graph at most once.; (b) Not one-to-one — e.g. $y = 4$ meets the graph at $x = \pm 2$.

17 [Challenging] • Target: Reading function graphs • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Using the graph of $f(x) = x^2 - 2$ above, estimate all x-values where $f(x) = 2$.

Setting $x^2 - 2 = 2$: $x^2 = 4 \Rightarrow x = \pm 2$, matching where the parabola crosses the horizontal line $y = 2$ on the graph.

18 [Challenging] • Target: Average rate of change from function notation • Cross-Lesson: 6.02 Polar Equations and Graphs

For $f(x) = x^2 + 1$, find the average rate of change of f from $x = 1$ to $x = 3$.

$\frac{f(3) - f(1)}{3 - 1} = \frac{10 - 2}{2} = 4$.

19 [Challenging] • Target: Difference quotients • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

For $f(x) = x^2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$.

$\frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h$.

20 [Challenging] • Target: Difference quotients • Cross-Lesson: 7.02 Logarithms and Their Properties

For $f(x) = 3x + 2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$, and explain why the result makes sense for a linear function.

$\frac{3(x+h) + 2 - (3x + 2)}{h} = \frac{3h}{h} = 3$. This makes sense because a linear function has constant slope 3 everywhere, so its average rate of change over any interval is always exactly that slope.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating Functions and Function Notation

Given $f(x) = 3x^2 - 4x + 1$, evaluate $f(-2)$.

a. 21 ✓ **b.** 5

c. -3 **d.** 17

$f(-2) = 3(-2)^2 - 4(-2) + 1 = 3(4) + 8 + 1 = 12 + 8 + 1 = 21$.

22 [Foundational] • Target: Evaluating Functions and Function Notation

If $g(x) = \frac{2x+5}{x-1}$, find $g(3a+1)$.

a. $\frac{6a+7}{3a}$ ✓ **b.** $\frac{6a+5}{3a+1}$

c. $\frac{6a+2}{3a}$ **d.** $\frac{2a+5}{a}$

$g(3a+1) = \frac{2(3a+1)+5}{(3a+1)-1} = \frac{6a+2+5}{3a} = \frac{6a+7}{3a}$.

23 [Foundational] • Target: Evaluating Functions and Function Notation

Find all values of x for which $f(x) = x^2 - 6x$ equals -8.

a. $x = 2$ **and** $x = 4$ ✓ **b.** $x = -2$ and $x = -4$

c. $x = 1$ and $x = 8$ **d.** $x = 0$ and $x = 6$

$x^2 - 6x = -8 \Rightarrow x^2 - 6x + 8 = 0 \Rightarrow (x-2)(x-4) = 0 \Rightarrow x = 2$ or $x = 4$.

24 [Foundational] • Target: The Vertical Line Test and Relations

Which of the following equations defines y as a function of x?

a. $y = 2x^3 - 5$ ✓ **b.** $x^2 + y^2 = 16$

c. $x = y^2 + 1$ **d.** $|y| = 3x$

$y = 2x^3 - 5$ assigns exactly one unique output y for each input x. Equations with y^2 or $|y|$ can produce two outputs for a single x.

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25 [Foundational] • Target: The Vertical Line Test and Relations

What does the Vertical Line Test determine about a graph?

a. Whether the graph represents a function ✓

b. Whether the function is one-to-one

c. The domain of the function

d. The slope of the curve

If every vertical line intersects a graph in at most one point, then the graph represents y as a function of x.

26 [Foundational] • Target: The Vertical Line Test and Relations

Which set of ordered pairs represents a function?

a. $\{(1,2),(2,3),(3,4),(4,4)\}$ ✓

b. $\{(1,2),(1,3),(2,4)\}$

c. $\{(0,5),(0,-5),(1,2)\}$

d. $\{(3,1),(3,2),(3,3)\}$

In $\{(1,2),(2,3),(3,4),(4,4)\}$, every first coordinate (x) is unique and paired with exactly one second coordinate (y).

27 [Foundational] • Target: Operations and Difference Quotients

Simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x)=4x-7$.

a. 4 ✓

b. $4h$

c. $4x+4h-7$

d. 0

$\frac{[4(x+h)-7]-[4x-7]}{h}=\frac{4x+4h-7-4x+7}{h}=\frac{4h}{h}=4.$

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28 [Foundational] • Target: Operations and Difference Quotients

Simplify $\frac{f(x+h)-f(x)}{h}$ for $f(x)=2x^2+3$.

a. $4x+2h$ ✓

b. $4x$

c. $2x+2h$

d. $4x+h$

$\frac{2(x+h)^2+3-(2x^2+3)}{h}=\frac{2(x^2+2xh+h^2)-2x^2}{h}=\frac{4xh+2h^2}{h}=4x+2h.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Operations and Difference Quotients

If $f(x)=x^2-1$ and $g(x)=2x+3$, find $(f\cdot g)(2)$.

a. 21 ✓

b. 15

c. 18

d. 28

$f(2)=2^2-1=3$. $g(2)=2(2)+3=7$. $(f\cdot g)(2)=f(2)\cdot g(2)=3\cdot 7=21$.

30 [Intermediate] • Target: Function Composition

Given $f(x)=2x+1$ and $g(x)=x^2-3$, find $(f\circ g)(x)$.

a. $2x^2-5$ ✓

b. $4x^2+4x-2$

c. $2x^2-3$

d. $2x^2+1$

$(f\circ g)(x)=f(g(x))=2(x^2-3)+1=2x^2-6+1=2x^2-5$.

31 [Intermediate] • Target: Function Composition

For the same functions $f(x)=2x+1$ and $g(x)=x^2-3$, evaluate $(g\circ f)(2)$.

a. 22 ✓

b. 25

c. 17

d. 3

$f(2)=2(2)+1=5$. Then $g(f(2))=g(5)=5^2-3=25-3=22$.

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32 [Intermediate] • Target: Function Composition

If $h(x) = \sqrt{2x^3 - 5}$, find functions $f(x)$ and $g(x)$ such that $h(x) = (f \circ g)(x)$.

a. $f(x) = \sqrt{x}, g(x) = 2x^3 - 5$ ✓

b. $f(x) = 2x^3 - 5, g(x) = \sqrt{x}$

c. $f(x) = x^3, g(x) = \sqrt{2x - 5}$

d. $f(x) = 2\sqrt{x}, g(x) = x^3 - 5$

Outer function is the square root: $f(u) = \sqrt{u}$. Inner function is the cubic polynomial: $g(x) = 2x^3 - 5$.

33 [Intermediate] • Target: Function Composition

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \geq 3$. Find $f(3)$.

a. -6

b. -4

c. 9

d. **-5** ✓

Since $x = 3 \geq 3$, use $f(x) = -2x + 1$: $f(3) = -2(3) + 1 = -5$.

34 [Intermediate] • Target: Function notation

A function is defined by $f(x) = x^2$ for $x < 1$ and $f(x) = 2x + 1$ for $x \geq 1$. Find $f(1)$.

a. 4

b. 1

c. **3** ✓

d. 2

Since $x = 1 \geq 1$, use $f(x) = 2x + 1$: $f(1) = 2(1) + 1 = 3$.

35 [Intermediate] • Target: Identifying functions

Determine whether $f(x) = 2x^4 - 5x^2 + 3$ is even, odd, or neither.

a. **Even** ✓

b. Odd

c. Neither

d. Cannot be determined

Every term has even degree, so $f(-x) = f(x)$: even.

36 [Intermediate] • Target: Combining functions

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. **Odd** ✓

c. Neither

d. Cannot be determined

Every term has odd degree, so $f(-x) = -f(x)$: odd.

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37 [Intermediate] • Target: Piecewise-defined functions

Determine whether $f(x) = x^5 - x^3$ is even, odd, or neither.

a. Even

b. **Odd ✓**

c. Neither

d. Cannot be determined

Every term has odd degree, so $f(-x) = -f(x)$: odd.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Even, odd, and one-to-one functions • Cross-Lesson: 6.02 Polar Equations and Graphs

A piecewise function is defined by $f(x) = 2x - 2$ for $x < -1$ and $f(x) = 3x + k$ for $x \geq -1$. Find k so that f is continuous at $x = -1$.

a. **-1 ✓**

b. -4

c. -3

d. 1

Continuity: $2(-1) - 2 = 3(-1) + k \Rightarrow k = -1$.

39 [Challenging] • Target: Reading function graphs • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A piecewise function is defined by $f(x) = -2x + 3$ for $x < -1$ and $f(x) = 4x + k$ for $x \geq -1$. Find k so that f is continuous at $x = -1$.

a. 11

b. -4

c. **9 ✓**

d. 5

Continuity: $-2(-1) + 3 = 4(-1) + k \Rightarrow k = 9$.

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40 [Challenging] • Target: Average rate of change from function notation • Cross-Lesson: 7.02 Logarithms and Their Properties

A piecewise function is defined by $f(x) = x + 6$ for $x < -1$ and $f(x) = -2x + k$ for $x \geq -1$. Find k so that f is continuous at $x = -1$.

a. 5

b. 2

c. 4

d. **3** ✓

Continuity: $1(-1) + 6 = -2(-1) + k \Rightarrow k = 3$.

41 [Challenging] • Target: Difference quotients • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A piecewise function is defined by $f(x) = -x + 5$ for $x < 2$ and $f(x) = 2x + k$ for $x \geq 2$. Find k so that f is continuous at $x = 2$.

a. 4

b. 3

c. **-1** ✓

d. 1

Continuity: $-1(2) + 5 = 2(2) + k \Rightarrow k = -1$.

42 [Challenging] • Target: Inverse functions (introduction) • Cross-Lesson: 7.04 Applications: Growth and Decay

A piecewise function is defined by $f(x) = 2x + 4$ for $x < -2$ and $f(x) = x + k$ for $x \geq -2$. Find k so that f is continuous at $x = -2$.

a. **2** ✓

b. 4

c. -2

d. 0

Continuity: $2(-2) + 4 = 1(-2) + k \Rightarrow k = 2$.

43 [Challenging] • Target: Evaluating Functions and Function Notation • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A piecewise function is defined by $f(x) = 2x + 1$ for $x < 3$ and $f(x) = -x + k$ for $x \geq 3$. Find k so that f is continuous at $x = 3$.

a. 7

b. **10** ✓

c. 12

d. -3

Continuity: $2(3) + 1 = -1(3) + k \Rightarrow k = 10$.

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44 [Challenging] • Target: The Vertical Line Test and Relations • Cross-Lesson: 8.02 Graphing Rational Functions

A piecewise function is defined by $f(x) = 2x + 5$ for $x < -3$ and $f(x) = 3x + k$ for $x \geq -3$. Find k so that f is continuous at $x = -3$.

a. -9

b. **8 ✓**

c. -1

d. 10

Continuity: $2(-3) + 5 = 3(-3) + k \Rightarrow k = 8$.

45 [Challenging] • Target: Operations and Difference Quotients • Cross-Lesson: 8.03 Introduction to Limits

A piecewise function is defined by $f(x) = -3x + 1$ for $x < 2$ and $f(x) = x + k$ for $x \geq 2$. Find k so that f is continuous at $x = 2$.

a. 2

b. -5

c. **-7 ✓**

d. 3

Continuity: $-3(2) + 1 = 1(2) + k \Rightarrow k = -7$.

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